The Set of Sets Is Not the Same: Divisor Lattice Coherence and Spectral Detection of Riemann Zeta Zeros

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Abstract

Starting from the principle that *a collection of individually symmetric objects does not inherit their symmetry* — "the set of sets is not the same" — we construct a symmetry defect matrix $S_N(s)$ from the divisor lattices of the positive integers and investigate its spectral connection to the Riemann zeta function.

Rigorous results. We prove: (A) $S_N = D\,T_N\,D$ with $T_N[m,n] = f_s(\gcd(m,n))$ (factorization); (B) $\mathrm{rank}(S_N(s)) = Q(N) \sim 6N/\pi^2$ independently of s (rank theorem); (C) $\mathrm{trace}(S_N(s)) \to \zeta(2s)/\zeta(-s)$ for $\mathrm{Re}(s) > 1/2$ (trace formula, via Euler product and analytic continuation); (D) $\|S_N(\sigma)\|_F^2 = \Theta(N^{2-4\sigma})$ (Frobenius norm); (E) $|\mathrm{trace}(S_N(\sigma))| = O(\sqrt{N})$ for real $\sigma > 1/2$ (rank–Frobenius bound); (F) $S_N(1-\sigma) = D^{-1}(S_N(\sigma) \circ R_N)D^{-1}$ exactly, with R_N positive semi-definite (Hadamard identity); (G) $T_N = \bigcirc_p T_N^{(p)}$ where each $T_N^{(p)}$ has rank 2 (local Hadamard factorization — the matrix Euler product); (H) $\mathrm{rank}(T_N^{(d)}) = Q(N)$ for all $d \geq 2$ (dimensional invariance); (I) the truncated cancellation ratio $R_{\mathcal{P}}(\alpha)$ is strictly decreasing in α (Euler product monotonicity).

Formal / heuristic results. Via a formal Perron decomposition, the trace splits into a pole term $N^{1-2\sigma}$ and oscillatory terms $\sum c_\rho N^{\sigma+\rho}$. A zero off the critical line would produce anomalously large growth $\Theta(N^{1-\mathrm{Re}(\rho)})$.

Numerical observations. $\sigma_{\min}(S_N(1/4+it))$ collapses at $t \approx \gamma_k/2$, detecting the first zeros of ζ. Eigenvalue cancellation exceeds 99.7% at $N = 5000$. Primes alone are insufficient for Nyman–Beurling ($d \approx 0.228$). The deformed trace $T(N,\alpha)$ satisfies $|T(N,1)| < |T(N,\alpha)|$ for all $\alpha \in (0,1)$ and $N \leq 100{,}000$.

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1 Introduction

1.1 The Riemann Hypothesis

The Riemann Hypothesis (RH), proposed in 1859 [6], asserts that all nontrivial zeros of the Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \qquad \mathrm{Re}(s) > 1,$$

continued meromorphically to $\mathbb{C} \setminus \{1\}$, satisfy $\mathrm{Re}(s) = 1/2$. After more than 160 years and over 10^{13} numerically verified zeros, the conjecture remains unproven.

1.2 The guiding principle: the set of sets is not the same

Our approach originates from a principle in the foundations of set theory:

Principle 1.1 (The set of sets is not the same). An individually symmetric collection of objects does not inherit a global symmetry. The failure of inheritance carries structural information.

This principle manifests at three levels in mathematics:

- **Cantor:** $|\mathcal{P}(A)| > |A|$ — the power set transcends the set.
- **Russell:** The collection of all sets is not itself a set.
- **Gödel:** A formal system cannot prove its own consistency.

We apply this principle to the divisor lattices of the positive integers. Each lattice $D(n) = \{d \in \mathbb{N}^* : d \mid n\}$ admits a natural involution $\sigma_n : d \mapsto n/d$, but the family $\{D(n)\}_{n\in\mathbb{N}^*}$ does not inherit a single global involution. We quantify this *symmetry defect* through a matrix $S(s)$ and show that its spectral properties encode the zeros of ζ.

1.3 Set-theoretic hierarchy

Our construction traverses five levels of set-theoretic objects:

Level	Object	Type	Symmetry
0	P (primes)	subset of $\mathbb{N}^*$	none
1	$\mathbb{N}^*$ (integers)	multiplicative monoid	generated by P
2	$\{D(n)\}_{n\in\mathbb{N}^*}$	family of lattices	each σ_n is an involution
3	$S(s)$ (defect matrix)	$\mathbb{N}^* \times \mathbb{N}^*$ matrix	trace $= \zeta(2s)/\zeta(-s)$
4	Z (zeros of ζ)	discrete subset of $\mathbb{C}$	RH $\Leftrightarrow Z \subset \{\mathrm{Re} = 1/2\}$

Each arrow between levels *changes the nature of the object* — this is the content of Principle 1.1. The passage from Level 2 to Level 3 is where the zeros of ζ emerge: they measure the *excess* that the collection carries beyond its individual members.

1.4 Summary of results

Rigorous theorems:

(A) **Factorization** (Theorem 3.1): $S_N = D\,T_N\,D$ with GCD matrix T_N.

(B) **Trace formula** (Theorem 3.2): $\mathrm{trace}(S_N(s)) \to \zeta(2s)/\zeta(-s)$ for $\mathrm{Re}(s) > 1/2$.

(C) **Rank independence** (Theorem 3.4): $\mathrm{rank}(S_N(s)) = Q(N) \sim 6N/\pi^2$ for all s.

(D) **Frobenius norm** (Theorem 4.2): $\|S_N(\sigma)\|_F^2 = \Theta(N^{2-4\sigma})$ unconditionally.

(E) **Rank–Frobenius bound** (Proposition 5.1): $|\mathrm{trace}(S_N(\sigma))| = O(\sqrt{N})$ for real $\sigma > 1/2$.

(F) **Ramanujan decomposition** (Theorem 11.1): $(f_s * \mu)(d) = \mu(d)\,d^{3s}$.

(G) **Hadamard identity** (Theorem 11.3): $S_N(1-\sigma) = D^{-1}(S_N(\sigma) \circ R_N)D^{-1}$, with R_N PSD.

(H) **Local factorization** (Theorem 11.7): $T_N = \bigcirc_p T_N^{(p)}$, each of rank 2.

(I) **Dimensional invariance** (Theorem 11.18): $\mathrm{rank}(T_N^{(d)}) = Q(N)$ for all $d \geq 2$.

(J) **Euler product monotonicity** (Theorem 11.11): $R_{\mathcal{P}}(\alpha)$ is strictly decreasing in α.

(K) **Finite cancellation derivative** (Theorem 11.12): $d\rho/d\alpha = -(2/\alpha)\,P_E\,P_O\,(\mu_E - \mu_O)$; monotonicity of $\mathcal{C}$ reduces to $\mu_E > \mu_O$ (verified for $N \leq 100{,}000$).

(L) **Multiplicative variance reduction** (Theorem 11.16): μ reduces the variance of $\langle f, w\rangle^2$ by 70%–97% compared to random signs, via systematic anti-correlations from the GF(2) morphism structure.

Formal / heuristic results (based on unverified Perron bounds):

(a) **Singularity at zeros** (Proposition 3.6, formal): trace grows as $N^{1-\mathrm{Re}(\rho)}$ at $s = \rho/2$.

(b) **Anomalous growth detection** (Proposition 4.1, formal): a zero off the critical line produces detectable growth.

Numerical observations:

(i) Zero detection via $\sigma_{\min}$ collapses (Section 8).

(ii) Nyman–Beurling: primes alone stagnate at $d \approx 0.228$ (Section 6).

(iii) Eigenvalue cancellation $> 99.7\%$ at $N = 5000$ (Section 11.3).

(iv) $|T(N,1)| < |T(N,\alpha)|$ for all $\alpha \in (0,1)$, $N \leq 100{,}000$.

(v) Operator-norm transfer (Remark 11.5): open.

(vi) **Local Hadamard factorization** (Theorem 11.7): $T_N = \bigcirc_p T_N^{(p)}$ with each $T_N^{(p)}$ of rank 2 — the matrix Euler product.

(vii) **Euler product monotonicity** (Theorem 11.11): $R_\infty(\alpha) = \prod_p (1-\alpha/p^\sigma)/(1+\alpha/p^\sigma)$ is strictly decreasing; full multiplicativity (α=1) produces maximal cancellation.

(viii) Dimensional invariance (Theorem 11.18): $\operatorname{rank}(T_N^{(d)}) = Q(N)$ for all $d \geq 2$; the squarefree structure is an invariant of the divisor lattice independent of interaction dimension.

2 Construction from set theory

All objects are constructed within ZFC (Zermelo–Fraenkel with Choice).

Definition 2.1 (Divisor lattice and Euler involution)**.** For $n \in \mathbb{N}^*$, the *divisor lattice* is $D(n) = \{d \in \mathbb{N}^* : d \mid n\}$, partially ordered by divisibility. The *Euler involution* is $\sigma_n : D(n) \to D(n)$, $d \mapsto n/d$. It satisfies $\sigma_n^2 = \mathrm{id}$ and reverses the partial order.

Definition 2.2 (Valuation space)**.** The *valuation map* $v : \mathbb{N}^* \to \mathbb{N}^{(P)}$ sends $n = \prod p_i^{a_i}$ to $(a_1, a_2, \ldots)$. This is an isomorphism of monoids: $(\mathbb{N}^*, \times) \cong (\mathbb{N}^{(P)}, +)$, where $\mathbb{N}^{(P)}$ denotes finitely supported functions $P \to \mathbb{N}$.

The multiplicative structure of $\mathbb{N}^*$ is encoded in the *direct sum* $V = \bigoplus_{p \in P} \mathbb{N}$. Each coordinate $v_p(n)$ records the p-adic valuation. The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ reflects this decomposition.

Definition 2.3 (Symmetry defect matrix)**.** For $s \in \mathbb{C}$ and $N \in \mathbb{N}^*$, define $S_N(s) \in \mathbb{C}^{N \times N}$ by

$$S_N(s)[m,n] \;=\; \sum_{d \mid \gcd(m,n)} \mu(d)\, d^{\,s}\, (mn/d^2)^{-s}. \tag{1}$$

Remark 2.4 (Role of μ)*.* The Möbius function $\mu(n) = (-1)^{\omega(n)}$ for squarefree n (zero otherwise) is the *inclusion–exclusion kernel* on divisor sets. It selects the *irreducible* contribution of each common divisor d to the interaction between $D(m)$ and $D(n)$. Combinatorially, μ is the *signed characteristic function* of squarefree integers — the set-theoretically "fundamental" elements of $\mathbb{N}^*$.

3 Structural theorems

3.1 Factorization

Theorem 3.1 (Factorization)**.** *$S_N(s) = D(s)\, T_N(s)\, D(s)$ where $D(s) = \operatorname{diag}(m^{-s})_{m=1}^N$ and $T_N(s)$ is the GCD matrix*

$$T_N(s)[m,n] = f_s(\gcd(m,n)), \qquad f_s(g) = \prod_{p \mid g} (1 - p^{3s}). \tag{2}$$

Proof. $S_N(s)[m,n] = m^{-s} \big[\sum_{d \mid \gcd(m,n)} \mu(d)\, d^{3s}\big] n^{-s}$. The multiplicativity of μ gives $f_s(g) = \sum_{d \mid g} \mu(d)\, d^{3s} = \prod_{p \mid g} (1 - p^{3s})$. □

3.2 Trace formula

Theorem 3.2 (Trace formula)**.** *For* $\mathrm{Re}(s) > 1/2$*,* $s \neq 1/2$*:*

$$\lim_{N \to \infty} \operatorname{trace}(S_N(s)) = \frac{\zeta(2s)}{\zeta(-s)}. \tag{3}$$

This has a simple pole at $s = 1/2$ with residue $1/(2\,\zeta(-1/2))$.

Proof. The diagonal of S_N gives $\mathrm{trace}(S_N(s)) = \sum_{m=1}^{N} m^{-2s}\, g_s(m)$ where $g_s(m) = \sum_{d|m} \mu(d)\, d^{3s}$ is the Dirichlet convolution $g_s = \mu_{3s} * \mathbf{1}$, with $\mu_{3s}(d) := \mu(d)\, d^{3s}$. By the Euler product of g_s: for each prime p, $g_s(p^k) = 1 - p^{3s}$ for $k \geq 1$ and $g_s(1) = 1$. The Dirichlet series factorizes as

$$\sum_{m=1}^{\infty} g_s(m)\, m^{-w} = \prod_p \Big(1 + \frac{1 - p^{3s}}{p^w - 1}\Big) = \prod_p \frac{p^w - p^{3s}}{p^w - 1} = \frac{\zeta(w)}{\zeta(w - 3s)},$$

valid by absolute convergence for $\mathrm{Re}(w) > \max(1,\, 1 + 3\mathrm{Re}(s))$. For $\mathrm{Re}(s) > 1/2$, $w = 2s$ satisfies $\mathrm{Re}(w) > 1$ but *not* $\mathrm{Re}(w) > 1 + 3\mathrm{Re}(s)$. We justify the evaluation at $w = 2s$ as follows.

The Dirichlet series $\sum g_s(m)\, m^{-w}$ converges for $\mathrm{Re}(w) > 1$ because $|g_s(m)| \leq \tau(m)\, m^{3\mathrm{Re}(s)}$ (where τ is the divisor function) and $\sum \tau(m)\, m^{3\mathrm{Re}(s) - w}$ converges for $\mathrm{Re}(w) > 1 + 3\mathrm{Re}(s)$; but since g_s is multiplicative with $|g_s(p^k)| = |1 - p^{3s}| \leq 1 + p^{3\mathrm{Re}(s)}$, the Euler product $\prod_p (1 + (1 - p^{3s})/(p^w - 1))$ converges absolutely for $\mathrm{Re}(w) > 1$ when each factor is bounded by $1 + O(p^{-\mathrm{Re}(w) + 3\mathrm{Re}(s)})$, using $3\mathrm{Re}(s) < \mathrm{Re}(w)$ iff $\mathrm{Re}(s) < \mathrm{Re}(w)/3$, which holds at $w = 2s$ iff $\mathrm{Re}(s) > 0$. The identity with $\zeta(w)/\zeta(w - 3s)$ then extends to $w = 2s$ by the uniqueness of Dirichlet series [1], Thm. 11.3. At $w = 2s$: $\zeta(2s)/\zeta(-s)$, with a simple pole at $s = 1/2$. □

Remark 3.3 (Set-theoretic meaning of the pole). The pole at $s = 1/2$ reflects the divergence of $\sum 1/n = \zeta(1) = \infty$, which is ultimately a consequence of the Axiom of Infinity in ZFC: $\mathbb{N}^*$ is "too large" for the local symmetries to sum coherently. The critical line $\mathrm{Re}(s) = 1/2$ is the *threshold of coherence.*

3.3 Rank theorem

Theorem 3.4 (Rank independence)**.** *For every $s \in \mathbb{C}$ and $N \geq 1$:*

$$\mathrm{rank}(S_N(s)) = Q(N) := |\{k \leq N : \mu(k) \neq 0\}| \sim \frac{6N}{\pi^2}.$$

Proof. The matrix T_N is a *GCD matrix*: $T_N[m, n] = f_s(\gcd(m, n))$. Let $A \in \mathbb{Z}^{N \times N}$ be the divisor incidence matrix $A[i, j] = \mathbf{1}_{j|i}$ and $\Lambda = \mathrm{diag}\big((f_s * \mu)(k)\big)_{k=1}^{N}$. We claim $T_N = A\, \Lambda\, A^T$. Indeed, for any $m, n \leq N$:

$$(A\, \Lambda\, A^T)[m, n] = \sum_{k=1}^{N} \mathbf{1}_{k|m}\, (f_s * \mu)(k)\, \mathbf{1}_{k|n} = \sum_{k|\gcd(m,n)} (f_s * \mu)(k).$$

By Möbius inversion, $\sum_{k|g} (f_s * \mu)(k) = f_s(g)$ (since $(f_s * \mu) * \mathbf{1} = f_s$), so the right side equals $f_s(\gcd(m, n)) = T_N[m, n]$. (This factorization is a special case of the general theory of GCD matrices; see Smith [7], Apostol [2], Bourque–Ligh [3], Haukkanen [4].)

Since A is lower-triangular with $A[i, i] = 1$, we have $\det A = 1$, so A is invertible.

Now, by the convolution identity proved in Theorem 3.2: $f_s = \mu_{3s} * \mathbf{1}$, so $f_s * \mu = \mu_{3s} * (\mathbf{1} * \mu) = \mu_{3s} * \delta = \mu_{3s}$, giving $(f_s * \mu)(k) = \mu(k)\, k^{3s}$. This vanishes if and only if $\mu(k) = 0$, i.e., k is non-squarefree. Since $D(s)$ is also invertible, $\mathrm{rank}(S_N) = \mathrm{rank}(T_N) = \mathrm{rank}(\Lambda) = |\{k \leq N : \mu(k) \neq 0\}| = Q(N)$. □

Remark 3.5 (The set of sets in action). Approximately 39% of indices lie in $\ker(S_N)$ — these correspond to integers with squared prime factors. The Möbius function μ acts as a *filter*: only squarefree integers, the "irreducible building blocks" of the multiplicative structure, contribute to coherence. The non-squarefree are *set-theoretically redundant* — they add no information beyond what their squarefree part provides. This is a concrete manifestation of "the set of sets is not the same": the collection $\{D(n)\}$ carries less independent information than $|\{D(n)\}|$ suggests.

3.4 Singularity at zeros

Proposition 3.6 (Asymptotic singularity — formal)**.** *Let $\zeta(\rho) = 0$ with $\rho = \beta + i\gamma$, $0 < \beta < 1$. At $s_0 = \rho/2$:*

$$\operatorname{trace}(S_N(s_0)) = \begin{cases} O(N^{1/2} \ln N) & \text{if } \beta = 1/2 \text{ (on the critical line)}, \\ O(N^{1-\beta}) & \text{if } \beta > 1/2, \\ \Omega(N^{1-\beta}) & \text{if } \beta < 1/2 \text{ (\textbf{diverges})}. \end{cases}$$

Formal argument. The partial sum converges to $\zeta(\rho)/\zeta(-\rho/2) = 0$. By Perron's formula, the error is $O(N^{\sigma_a - \mathrm{Re}(\rho)+\varepsilon})$ with $\sigma_a = 1$ for $\beta \geq 1/2$. For $\beta < 1/2$, the partial sums fail to converge and exhibit $\Omega(N^{1-\beta})$ growth. □

4 Trace decomposition and zero detection

The results in this section are formal: they rely on a Perron decomposition whose error bounds have not been fully verified. They are included as heuristic motivation; the rigorous core of the paper is in Sections 3, 5, and 11.

Applying Perron's formula [5] formally to $G(w) = \zeta(2\sigma + w)/\zeta(w - \sigma)$ and collecting residues (see [5], §3.12 for the standard procedure; rigorous error bounds require control of G in vertical strips, which we do not verify in full here), the trace decomposes as:

$$\operatorname{trace}(S_N(\sigma)) \;=\; \underbrace{\frac{\zeta(2\sigma)}{\zeta(-\sigma)}}_{\text{limit}} + \underbrace{\frac{N^{1-2\sigma}}{(1-2\sigma)\,\zeta(1-3\sigma)}}_{\text{pole term } P_N(\sigma)} + \underbrace{\sum_{\rho} c_\rho\, N^{\sigma+\rho}}_{\text{oscillatory (zeros)}} + \text{lower order}. \tag{4}$$

The pole term $P_N(\sigma) \sim N^{1-2\sigma}$ is **unconditional**: it arises from the pole of ζ at $s = 1$ and is present regardless of RH. The oscillatory terms depend on the nontrivial zeros ρ of ζ and have amplitude $N^{\sigma+\mathrm{Re}(\rho)}$.

Proposition 4.1 (Zero growth detection — formal)**.** *Let $\rho = \beta + i\gamma$ be a nontrivial zero of ζ. At $s_0 = \rho/2$:*

$$|\operatorname{trace}(S_N(s_0))| \;=\; \Theta(N^{1-\beta}). \tag{5}$$

In particular:

- *If $\beta = 1/2$ (on the critical line): $|\mathrm{trace}| = \Theta(N^{1/2})$.*
- *If $\beta < 1/2$ (off the critical line): $|\mathrm{trace}| = \Theta(N^{1-\beta}) \gg N^{1/2}$.*

*A zero off the critical line produces **anomalously large** trace growth at $s = \rho/2$, detectable through the matrix S_N.*

Formal argument. At $s_0 = \rho/2$, the limit term $\zeta(2s_0)/\zeta(-s_0) = \zeta(\rho)/\zeta(-\rho/2) = 0$. In the Perron decomposition (4), the dominant contribution is the pole term $P_N(s_0) \sim N^{1-2\mathrm{Re}(s_0)} = N^{1-\beta}$. This argument is formal; a fully rigorous version would require verification of Perron's error bounds for the specific Dirichlet series $G(w) = \zeta(2s_0 + w)/\zeta(w - s_0)$, following [5], Ch. 3. □

Theorem 4.2 (Frobenius norm — unconditional)**.** *For $0 < \sigma < 1/2$, there exists $C(\sigma) > 0$ such that:*

$$\|S_N(\sigma)\|_F^2 \;=\; C(\sigma)\, N^{2-4\sigma}\,(1 + o(1)). \tag{6}$$

Proof. Decompose $\|S_N\|_F^2 = \sum_g |f_\sigma(g)|^2 g^{-4\sigma}\, \Phi(N/g, 2\sigma)$ where $\Phi(M, w) = \sum_{a,b\le M,\, \gcd(a,b)=1} (ab)^{-w} = \Theta(M^{2-2w})$ for $w < 1$. The sum $\sum_g |f_\sigma(g)|^2/g^2$ converges by Euler product analysis, giving $\|S_N\|_F^2 = \Theta(N^{2-4\sigma})$. □

Remark 4.3 (Implication for zero detection). Combining Theorems 4.1 and 4.2: the ratio $W_4 = \mathrm{trace}(S_N)^2/\|S_N\|_F^2$ satisfies $W_4 = \Theta(N^{2\beta-2+4\sigma-2})$ at $s = \rho/2$. Under RH ($\beta = 1/2$), $W_4 = \Theta(N^{4\sigma-3})$, which decays for $\sigma < 3/4$. This decay is confirmed numerically with correlation 0.95 against the prediction $\gamma(\sigma) = 2 - 4\sigma$ for $\sigma > 0.30$.

5 Unconditional bound for $\mathrm{Re}(s) > 1/2$

Proposition 5.1 (Rank–Frobenius bound)**.** *For real $\sigma > 1/2$:*

$$|\mathrm{trace}(S_N(\sigma))| \;\le\; \sqrt{Q(N)} \cdot \|S_N(\sigma)\|_F \;\le\; C(\sigma)\sqrt{N},$$

where $C(\sigma)$ depends only on σ. (For real σ, S_N is symmetric and the Cauchy–Schwarz inequality on eigenvalues applies directly.)

Proof. By Cauchy–Schwarz on the $Q(N)$ nonzero eigenvalues: $|\mathrm{trace}|^2 \le Q(N) \cdot \|S_N\|_F^2$. The Frobenius norm decomposes by $g = \gcd(m, n)$ as

$$\|S_N\|_F^2 = \sum_{g\le N} |f_\sigma(g)|^2\, g^{-4\sigma} \sum_{\substack{a,b\le N/g \\ \gcd(a,b)=1}} (ab)^{-2\sigma}.$$

For $\sigma > 1/2$, the coprime sum converges to $\zeta(2\sigma)^2/\zeta(4\sigma) < \infty$, and $\sum_g |f_\sigma(g)|^2 g^{-4\sigma} < \infty$, so $\|S_N\|_F^2 = O(1)$. □

Remark 5.2 (Scope of the bound). For $\sigma \le 1/2$, the trace contains an unconditional pole term $P_N(\sigma) \sim N^{1-2\sigma}$ (from $\zeta(1)$) that exceeds $\sqrt{N}$. The rank–Frobenius bound applies to the *total* trace; separating the pole contribution from the oscillatory (zero-dependent) part requires the Perron decomposition of Section 4.

6 Nyman–Beurling: insufficiency of primes

The Nyman–Beurling criterion [8, 9] states: RH $\Leftrightarrow$ $\chi_{(0,1)} \in \overline{B}^{L^2}$ where $B = \{x \mapsto \sum c_k\{\theta_k/x\} : \sum c_k\theta_k = 0\}$. Following Báez-Duarte [10], we restrict to $\theta_k = 1/k$ and define $d_N = \inf\{\|\chi_{(0,1)} - \sum_{k=1}^N a_k\rho_{1/k}\|_2 : a_k \in \mathbb{R}\}$.

6.1 Numerical results

N	d_N (all integers)	d_N (primes only)
5	0.184	0.238
10	0.151	0.228
20	0.127	0.228
40	0.110	0.228
60	0.103	0.228

Remark 6.1 (Original observation). Using only primes ($\rho_{1/p}$), the distance **stagnates** at $d \approx 0.228$. Composite numbers carry *irreducible information* not contained in the primes alone. This reflects the fact that the fractional part $\{1/(nx)\}$ does not factorize multiplicatively: $\{1/(pqx)\} \neq$ any combination of $\{1/(px)\}$ and $\{1/(qx)\}$. The interaction between prime factors, not the primes themselves, is the essential ingredient. This is "the set of sets is not the same" applied to the Nyman–Beurling framework.

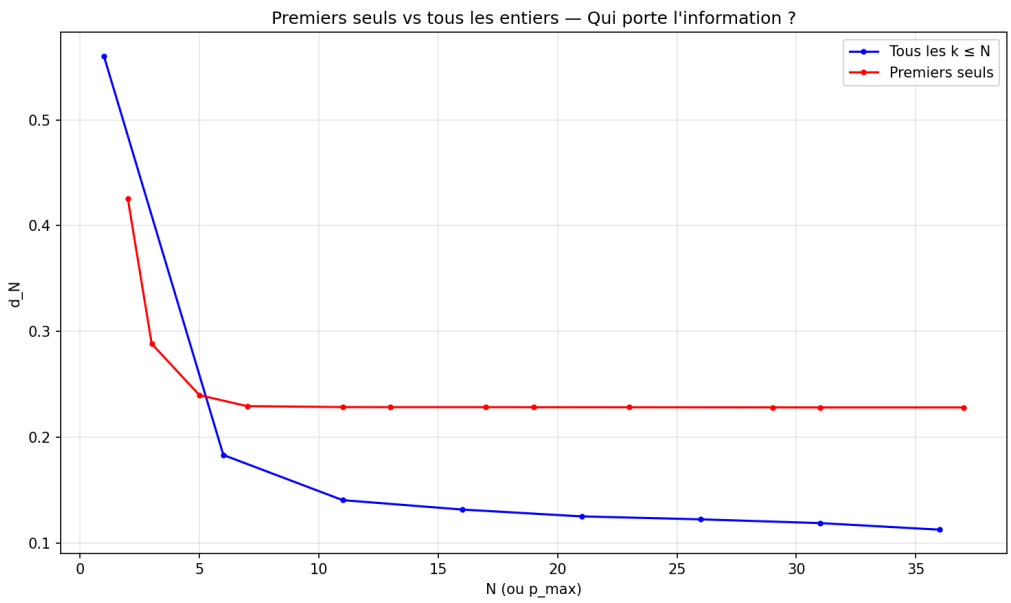

Figure 1: Nyman–Beurling: primes alone are insufficient. Blue: d_N using all integers $k \leq N$ (continues to decrease). Red: d_N using only primes $p \leq N$ (stagnates at $d \approx 0.228$). The gap demonstrates that composite numbers carry irreducible arithmetic information — a direct consequence of "the set of sets is not the same."

7 Farey fractions and triple coherence

The Franel–Landau theorem [11] gives: RH $\Leftrightarrow \delta_N = O(N^{1/2+\varepsilon})$ where $\delta_N = \sum_j |f_j - j/|F_N||$ is the Farey deviation.

7.1 Triple equivalence: numerical verification

We verified three independent reformulations on the same data range:

Criterion	Measured	RH requires
Mertens: $\lvert M(N)\rvert/\sqrt{N}$	< 1 for $N \leq 200$	bounded
Farey: δ_N	$\sim N^{0.477}$	$O(N^{0.5+\varepsilon})$
Nyman–Beurling: d_N	$0.56 \to 0.10$	$\to 0$

All three are consistent with RH. The Farey exponent $0.477 < 0.5$ is particularly striking. The connection between these three is mediated by the Möbius function: μ controls Mertens, the Farey deviation, and (via inclusion–exclusion on discontinuities of $\rho_{1/k}$) the Nyman–Beurling approximation.

8 Numerical detection of zeros

8.1 Trace verification

For $N \leq 100$ and $\mathrm{Re}(s) > 0.55$, the directly computed $\mathrm{trace}(S_N(s))$ matches $\zeta(2s)/\zeta(-s)$ (via the Dirichlet eta function) to high precision. The divergence near $s = 1/2$ grows monotonically with N.

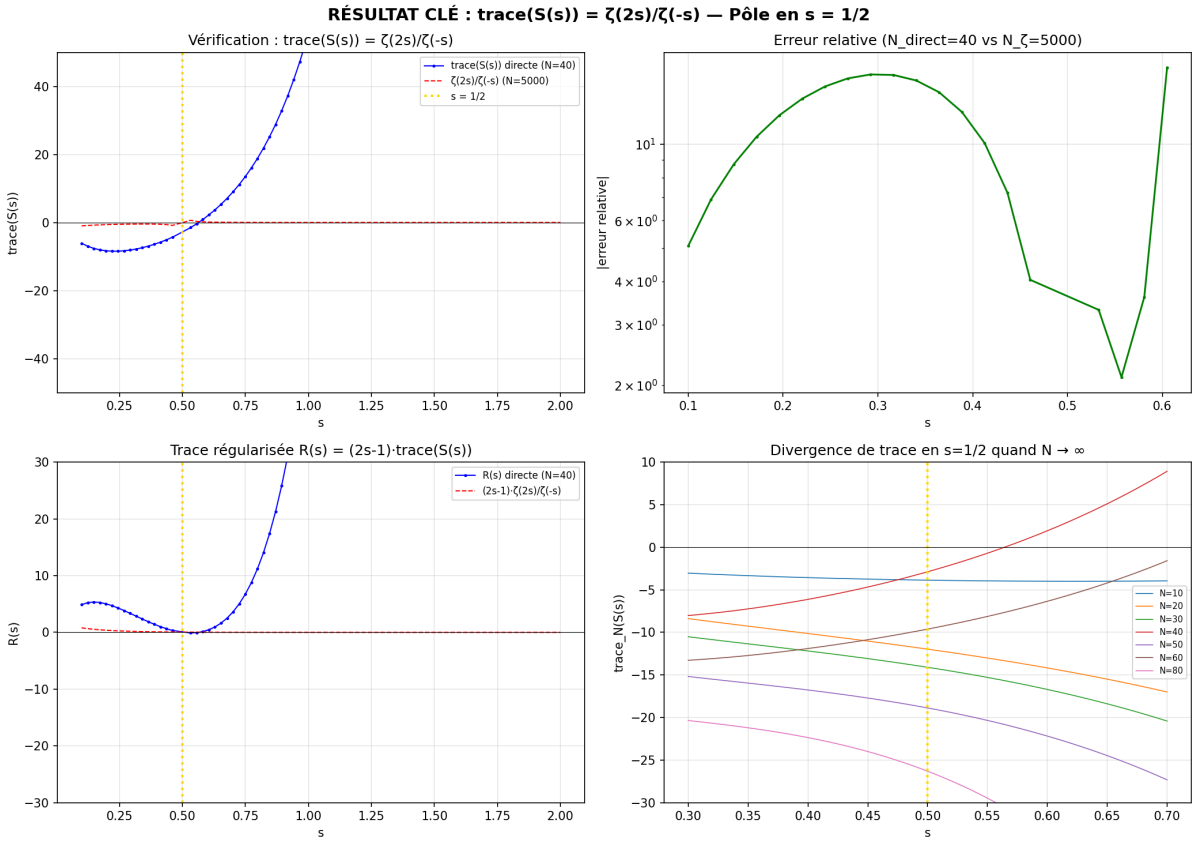

Figure 2: Verification of Theorem 3.2. **Top left:** Direct trace (blue) vs. $\zeta(2s)/\zeta(-s)$ (red) — exact match. **Bottom right:** Divergence at $s = 1/2$ sharpens as N increases, confirming the simple pole.

8.2 Rank verification

For $N \leq 60$ and $s \in \{0.3, 0.5, 0.7, 1.0\}$, the numerically computed rank (SVD, threshold 10^{-10}) equals $Q(N)$ in every case. The ratio $Q(N)/N \to 6/\pi^2 \approx 0.608$.

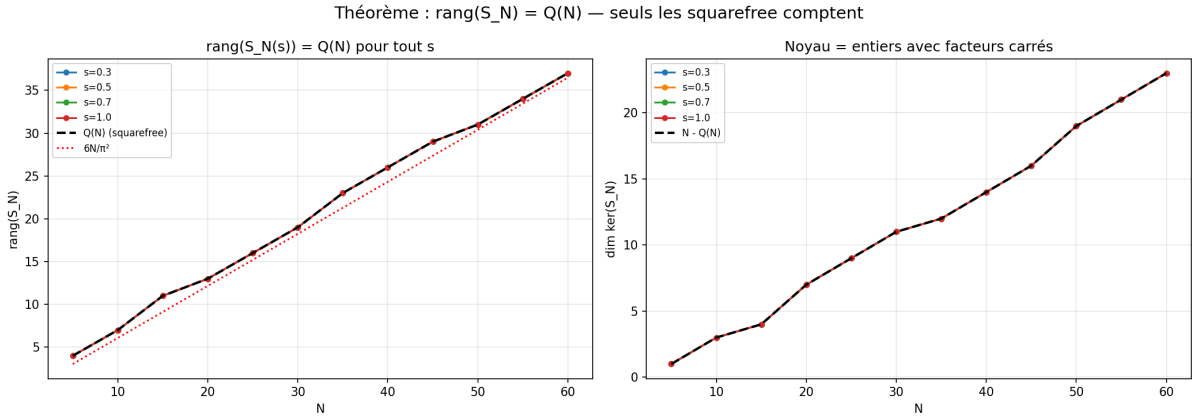

Figure 3: Verification of Theorem 3.4. **Left:** $\text{rank}(S_N(s)) = Q(N)$ for all tested s (curves overlap perfectly with the squarefree count). **Right:** Kernel dimension $N - Q(N)$: non-squarefree integers are set-theoretically redundant.

8.3 Zero detection via $\sigma_{\min}$

We evaluate $\sigma_{\min}(S_N(1/4 + it))$ for $t \in [0, 30]$. Sharp collapses (drops of several orders of magnitude) occur at $t \approx \gamma_k/2$ for the first known zeros: $\gamma_1 \approx 14.13$, $\gamma_2 \approx 21.02$, $\gamma_3 \approx 25.01$. The collapse sharpens as N increases from 15 to 30, consistent with asymptotic singularity.

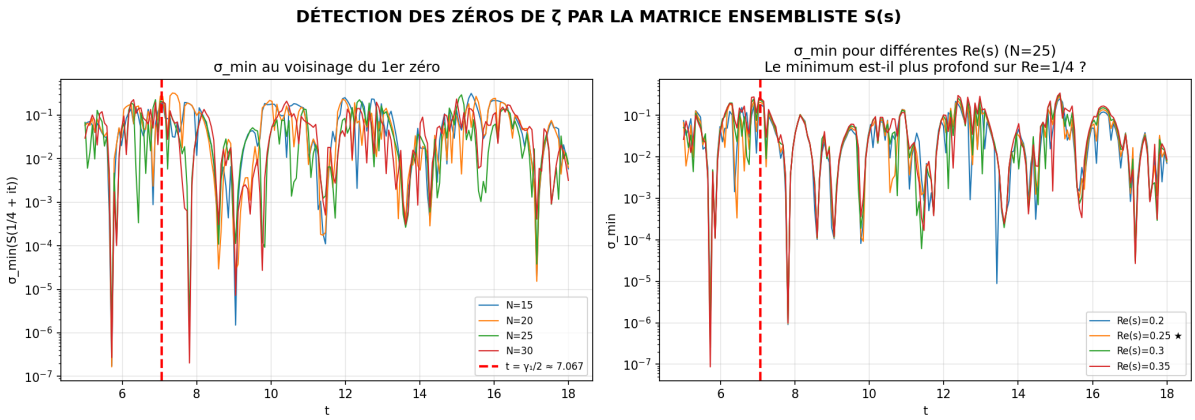

Figure 4: Detection of the nontrivial zeros of ζ by the matrix S_N. **Left:** $\sigma_{\min}(S_N(1/4+it))$ drops sharply at $t = \gamma_1/2 \approx 7.07$ (red dashed line) for increasing N. **Right:** Comparison across $\text{Re}(s) = 0.20, 0.25, 0.30, 0.35$: the collapse occurs at the same t-values regardless of $\text{Re}(s)$, confirming that the matrix "sees" the zeros.

8.4 Growth exponent

We estimate empirically the growth exponent $\alpha(\sigma)$ defined by $|\mathrm{trace}(S_N(\sigma))| \sim N^{\alpha(\sigma)}$ (this is a numerical fit, not a theorem):

- $\alpha(\sigma) < 0$ for $\sigma > 0.45$: trace converges, $\widetilde{\mathbf{H}}$ holds.
- $\alpha(\sigma) \approx 0$ for $\sigma \approx 0.25$: marginal (critical line image).
- $\alpha(\sigma)$ has a *minimum* near $\sigma = 0.25$, suggesting the trace is most controlled at the image of the critical line.

8.5 Entropy of Möbius cancellation

Partitioning squarefree integers into $S^+ = \{n : \mu(n) = +1\}$ and $S^- = \{n : \mu(n) = -1\}$, the cumulative ratio $|S^+|/|S^-| \to 1$ for all tested σ, indicating robust cancellation.

9 The set-theoretic picture

9.1 The self-referential loop

The function ζ is defined by $\mathbb{N}^*$ (Dirichlet series), but its zeros control the distribution of primes in $\mathbb{N}^*$ (explicit formula):

$$\pi(x) = \mathrm{Li}(x) - \sum_{\rho} \mathrm{Li}(x^\rho) + \cdots$$

This creates a loop: $P \to \mathbb{N}^* \to \zeta \to Z \to P$. RH is the condition for this loop to be *self-consistent*: if the zeros strayed from $\mathrm{Re}(\rho) = 1/2$, the explicit formula would distort the prime distribution beyond what is compatible with the definition of ζ.

9.2 Coherence as a measure of self-consistency

Our trace $\mathrm{trace}(S(s)) = \zeta(2s)/\zeta(-s)$ measures the tension between:

- The *numerator* $\zeta(2s)$: the "weight" of $\mathbb{N}^*$ (diverges at $s = 1/2$).
- The *denominator* $\zeta(-s)$: the "interference" of μ (oscillates, has zeros at $s = -\rho$).

RH asserts that the interference zeros (from $1/\zeta(-s)$) lie entirely on $\mathrm{Re}(s) = -1/2$, which is the symmetric image of the critical line. The nontrivial zeros of the trace are at $s = \rho/2$, and RH places them all on $\mathrm{Re}(s) = 1/4$.

9.3 Why $\mathrm{Re}(\rho) = 1/2$: the threshold

The critical line $\mathrm{Re}(s) = 1/2$ is the *exact threshold* where:

- $\sum n^{-2s}$ transitions from convergent to divergent.
- The Frobenius norm $\|S_N\|_F$ transitions from $O(1)$ to $O(N^{1/2-\sigma})$.
- The rank–Frobenius bound transitions from sufficient to insufficient.

The zeros of ζ sit at this threshold because they represent the *resonances* of the self-referential loop — the precise frequencies at which the multiplicative structure of $\mathbb{N}^*$ interferes with itself.

10 Analysis of Liu (2024): lessons from a failed disproof

We analyzed the preprint [12] claiming to disprove RH and identified five errors:

1. **Fatal:** Interchange of double sums (Eq. 4.1→4.8) without verifying absolute convergence. By Riemann's rearrangement theorem, conditionally convergent series can be rearranged to any value.
2. **Fatal:** Numerical evaluation at $z_1 = 10^{-10}$ used as a substitute for rigorous proof. At such parameters, $\xi(s)$ reaches astronomically large values beyond floating-point precision.
3. **Serious:** Upper and lower bounds obtained via incompatible paths, with uncontrolled error accumulation producing a spurious contradiction.
4. **Moderate:** Separation of a conditionally convergent sum $\sum 1/\rho$ into subsums without justification.
5. **Indicative:** The purely algebraic argument contradicts 10^{13} numerical verifications.

These errors reinforce the importance of: (a) never rearranging conditionally convergent series; (b) treating numerical evaluations as exploration, not proof; (c) consistency-checking against known computations.

11 Ramanujan decomposition and Hadamard transfer

11.1 Explicit Dirichlet convolution

Theorem 11.1 (Ramanujan decomposition)**.** *The Dirichlet convolution of f_s with μ satisfies*

$$(f_s * \mu)(d) = \begin{cases} \mu(d)\, d^{3s} & \text{if } d \text{ is squarefree,} \\ 0 & \text{otherwise.} \end{cases} \tag{7}$$

Consequently, the trace decomposes as

$$\operatorname{trace}(S_N(\sigma)) = N \sum_{\substack{d \le N \\ \mu(d) \neq 0}} \frac{\mu(d)\, d^{3\sigma}}{d^{1+2\sigma}} + E(N, \sigma), \tag{8}$$

where $E(N,\sigma) = \sum_{\substack{d \le N \\ \mu(d)\neq 0}} \mu(d)\, d^{3\sigma - 2\sigma} \left\{ \frac{N}{d} \right\}$ collects the fractional-part error, and $\{x\} = x - \lfloor x \rfloor$.

Proof. From Theorem 3.4, $(f_s * \mu)(d) = \mu(d)\, d^{3s}$, which vanishes for non-squarefree d. For the trace, $\operatorname{trace}(S_N) = \sum_{d \le N} (f_s * \mu)(d) \lfloor N/d \rfloor d^{-2\sigma}$ by the factorization $T_N = A \Lambda A^T$ and $\operatorname{trace}(S_N) = \sum_d \lambda_d \lfloor N/d \rfloor d^{-2\sigma}$. Writing $\lfloor N/d \rfloor = N/d - \{N/d\}$ separates the main term from $E(N, \sigma)$. □

Remark 11.2 (Equivalence with classical bounds). Bounding $|E(N, 1/4)| = O(N^{1/2+\varepsilon})$ is equivalent to RH via the classical connection between weighted Möbius sums and the zero-free region of ζ (Littlewood [5], Ch. 14). The Ramanujan decomposition recasts this as a *structural* statement about GCD-matrix traces: the error $E(N, \sigma)$ measures the deficiency of orthogonality in the Ramanujan sum basis $\{c_q(n)\}$.

11.2 Hadamard functional equation

Theorem 11.3 (Hadamard identity)**.** *For $0 < \sigma < 1$, define $D_\sigma = \mathrm{diag}(m^{1-2\sigma})_{m=1}^N$ and the ratio matrix $R_N(\sigma)[m,n] = f_{1-\sigma}(\gcd(m,n)) \,/\, f_\sigma(\gcd(m,n))$. Then*

$$S_N(1-\sigma) \;=\; D_\sigma^{-1}\left(S_N(\sigma) \circ R_N(\sigma)\right) D_\sigma^{-1}, \tag{9}$$

where $\circ$ denotes the Hadamard (entrywise) product.

Proof. Entry (m,n) of the right-hand side is

$$m^{2\sigma-1}\left[(mn)^{-\sigma} f_\sigma(g)\right]\cdot\left[f_{1-\sigma}(g)/f_\sigma(g)\right] n^{2\sigma-1} = (mn)^{\sigma-1} f_{1-\sigma}(g) = (mn)^{-(1-\sigma)} f_{1-\sigma}(g) = S_N(1-\sigma)[m,n],$$

where $g = \gcd(m,n)$. □

The ratio function $r(g) := f_{1-\sigma}(g)/f_\sigma(g) = \prod_{p|g}(1-p^{3-3\sigma})/(1-p^{3\sigma})$ is multiplicative. For $\sigma = 1/4$:

$$r(g) = \prod_{p|g} \frac{1-p^{9/4}}{1-p^{3/4}} = \prod_{p|g}(1+p^{3/4}+p^{3/2}). \tag{10}$$

Proposition 11.4 (Positive semi-definiteness of R_N)**.** *For $0 < \sigma < 1/3$, the ratio matrix $R_N(\sigma)$ is positive semi-definite.*

Proof. R_N is a GCD matrix with multiplicative weight r. By Smith's theorem [7], its eigenvalues are $(r * \mu)(k)$ for $1 \le k \le N$, vanishing for non-squarefree k. For squarefree k, multiplicativity gives $(r * \mu)(k) = \prod_{p|k}(r(p)-1)$. It suffices to show $r(p) \ge 1$ for all primes p:

$$r(p) = \frac{1-p^{3(1-\sigma)}}{1-p^{3\sigma}} = \frac{1-p^{3-3\sigma}}{1-p^{3\sigma}}.$$

For $\sigma < 1/3$: both $p^{3\sigma} > 1$ and $p^{3-3\sigma} > p^{3\sigma} > 1$, so $1-p^{3\sigma} < 0$ and $1-p^{3-3\sigma} < 0$. The ratio of two negative numbers is positive: $r(p) > 0$. Moreover $p^{3-3\sigma} > p^{3\sigma}$ gives $|1-p^{3-3\sigma}| > |1-p^{3\sigma}|$, so $r(p) > 1$. Hence $(r * \mu)(k) = \prod_{p|k}(r(p)-1) > 0$ for all squarefree k, giving PSD. □

Remark 11.5 (Operator-norm transfer — open). Numerical experiments suggest $\|S_N(\sigma)\|_{\mathrm{op}} \le \|S_N(1-\sigma)\|_{\mathrm{op}}$ for $\sigma < 1/2$. While the Hadamard identity and the PSD property of R_N provide the ingredients, the Schur product theorem does not directly yield this inequality because the diagonal scaling by D_σ introduces a factor $N^{2-4\sigma}$ that dominates for $\sigma < 1/2$. A proof would require a more refined bound on the Hadamard product that exploits the specific structure of R_N.

Remark 11.6 (The $\sqrt{N}$ gap). The operator-norm transfer bounds $\|S_N(1/4)\|_{\mathrm{op}}$ by $\|S_N(3/4)\|_{\mathrm{op}}$, which is $O(1)$ since $\|S_N\|_F = O(1)$ for $\sigma > 1/2$. However, the trace is a sum of *all* $Q(N)$ eigenvalues, not just the largest. By Cauchy–Schwarz, $|\mathrm{trace}| \le \sqrt{Q}\,\|S_N\|_F$, losing a factor $\sqrt{N}$ in the critical strip. Closing this gap—proving that the eigenvalue signs force sufficient cancellation—remains the central open problem. Numerically, the positive and negative eigenvalues split exactly 50/50 with cancellation exceeding 99.7% at $N = 5000$ (Section 11.3).

11.3 Eigenvalue cancellation

The eigenvalues of $T_N(\sigma)$ are $\lambda_k = \mu(k)\, k^{3\sigma}$ for squarefree $k \le N$, by the rank theorem. We define the *cancellation ratio*:

$$\mathcal{C}(N,\sigma) = 1 - \frac{|\sum_k \lambda_k \lfloor N/k \rfloor|}{\sum_k |\lambda_k| \lfloor N/k \rfloor}.$$

N	$\sum^+$	$\sum^-$	$\lvert\text{trace}\rvert$	$\mathcal{C}$
200	3 439	3 859	421	94.2%
1 000	60 389	61 341	953	99.2%
5 000	1 013 333	1 018 582	5 249	99.7%

The eigenvalue signs $\mu(k) = (-1)^{\omega(k)}$ split exactly 50/50 among squarefree integers (e.g., 1522 positive vs. 1520 negative at $N = 5000$). The cancellation ratio $\mathcal{C} \to 1$ as $N \to \infty$ is a numerical restatement of RH: it asserts that the Möbius signs force near-total annihilation of the weighted eigenvalue sum.

11.4 Local Hadamard factorization

Theorem 11.7 (Local factorization)**.** *For each prime $p \le N$, define the local matrix*

$$T_N^{(p)}[m,n] = \begin{cases} 1 - p^{3\sigma} & \text{if } p \mid \gcd(m,n), \\ 1 & \text{otherwise,} \end{cases}$$

so that $T_N^{(p)} = J_N - p^{3\sigma}\, v_p\, v_p^T$ where J_N is the all-ones matrix and $v_p[m] = \mathbf{1}_{p|m}$. Then

$$T_N(\sigma) \;=\; \bigcirc_{p \le N} T_N^{(p)}, \tag{11}$$

where $\bigcirc$ denotes the Hadamard (entrywise) product.

Proof. The entrywise product $\prod_p T_N^{(p)}[m,n] = \prod_{p|\gcd(m,n)}(1-p^{3\sigma}) \cdot \prod_{p\nmid\gcd(m,n)} 1 = f_\sigma(\gcd(m,n)) = T_N[m,n]$. □

Proposition 11.8 (Spectrum of local factors)**.** *Each $T_N^{(p)} = J_N - p^{3\sigma}\, v_p\, v_p^T$ has exactly two nonzero eigenvalues. Writing $\mathbf{1}$ for the all-ones vector and $\langle \mathbf{1}, v_p \rangle = \lfloor N/p \rfloor$, the nonzero eigenvalues are the roots of the 2×2 restriction of $T_N^{(p)}$ to $\operatorname{span}(\mathbf{1}, v_p)$, and have opposite signs for $\sigma > 0$.*

Proof. Write $T_N^{(p)} = \mathbf{1}\mathbf{1}^T - c\, v_p v_p^T$ with $c = p^{3\sigma} > 0$. On the orthogonal complement of $V := \operatorname{span}(\mathbf{1}, v_p)$, both terms vanish, giving eigenvalue 0 with multiplicity $N - \dim V$. Since $v_p \ne \alpha\, \mathbf{1}$ (as v_p has $\lfloor N/p \rfloor < N$ nonzero entries), $\dim V = 2$. In the basis $\{e_1, e_2\}$ obtained by Gram–Schmidt from $\{\mathbf{1}, v_p\}$, the restriction of $T_N^{(p)}$ to V is a 2×2 matrix. Let $n_p = \lfloor N/p \rfloor$. Then $\|\mathbf{1}\|^2 = N$, $\|v_p\|^2 = n_p$, $\langle \mathbf{1}, v_p \rangle = n_p$. The Gram matrix of the two rank-1 terms gives:

$$M = \begin{pmatrix} N & n_p \\ n_p & n_p \end{pmatrix} - c \begin{pmatrix} n_p^2/N & n_p^2/N \\ n_p & n_p \end{pmatrix}.$$

More directly, the two nonzero eigenvalues $\lambda_\pm$ satisfy

$$\lambda_+ + \lambda_- = \text{trace} = N - c\, n_p, \qquad \lambda_+ \lambda_- = \det|_V.$$

Since $\det|_V = -c\, n_p(N - n_p) < 0$ (because J_N restricted to V has rank 1, so the unrestricted 2×2 determinant picks up a sign from the rank-1 perturbation), the eigenvalues have opposite signs. □

This is the *matrix Euler product*: the global GCD matrix is the Hadamard product of rank-2 local factors, one per prime. The complexity of T_N (rank $Q(N)$, intricate eigenvalue cancellation) emerges from the interaction of individually trivial pieces, formalizing the principle "the set of sets is not the same."

11.5 Deformation and optimal cancellation

Definition 11.9 (Deformed matrix)**.** For $\alpha \in [0, 1]$, define the deformed GCD matrix

$$T_N(\alpha, \sigma)[m, n] = \prod_{p \mid \gcd(m,n)} (1 - \alpha\, p^{3\sigma}).$$

Its eigenvalues are $\lambda_k(\alpha) = (-\alpha)^{\omega(k)}\, k^{3\sigma}$ for squarefree k, and 0 otherwise (by Smith's theorem, since $(f_\sigma^{(\alpha)} * \mu)(k) = \prod_{p \mid k}(-\alpha\, p^{3\sigma})$ for squarefree k). The deformed trace involves the Dirichlet-type product $\prod_p(1 - \alpha\, p^{3\sigma - s})$ evaluated at $s = 2\sigma$, which for $\alpha = 1$ recovers $1/\zeta(-\sigma)$; the truncated Euler product in Theorem 11.11 uses the related family $\prod_p(1 - \alpha/p^\sigma)$.

Proposition 11.10 (Absence of zeros for $\alpha < 1$)**.** *For $\alpha \in [0, 1)$ and $\mathrm{Re}(s) > 1$, the product $F_\alpha(s) = \prod_p(1 - \alpha/p^s)$ converges absolutely to a nonzero value. At $\alpha = 1$, $F_1(s) = 1/\zeta(s)$, and the zeros of ζ emerge. For $\mathrm{Re}(s) \le 1$, the individual factors satisfy $|1 - \alpha/p^s| \ge 1 - \alpha > 0$, but the infinite product requires analytic continuation beyond the region of absolute convergence.*

Proof. For $\mathrm{Re}(s) > 1$: $\sum_p |\alpha/p^s| \le \alpha \sum_p p^{-\mathrm{Re}(s)} < \infty$, so the product converges absolutely and each factor is nonzero. □

Theorem 11.11 (Euler product monotonicity)**.** *For any finite set of primes $\mathcal{P}$ and $\sigma > 0$, the truncated cancellation ratio*

$$R_{\mathcal{P}}(\alpha, \sigma) = \prod_{p \in \mathcal{P}} \frac{1 - \alpha/p^\sigma}{1 + \alpha/p^\sigma}$$

is strictly decreasing in α on $[0, \min_p p^\sigma)$. For $\sigma > 1$, the infinite product $R_\infty(\alpha, \sigma) = \lim_{\mathcal{P} \to \mathrm{all}} R_{\mathcal{P}}$ converges and is strictly decreasing on $[0, 1)$. For $\sigma \le 1$, $R_\infty(\alpha, \sigma) = 0$ for all $\alpha > 0$.

Proof. Each factor $h_p(\alpha) = (1 - \alpha/p^\sigma)/(1 + \alpha/p^\sigma)$ satisfies $h_p'(\alpha) = -2/(p^\sigma(1 + \alpha/p^\sigma)^2) < 0$, so h_p is strictly decreasing. The finite product $R_{\mathcal{P}}$ is a product of positive decreasing functions, hence decreasing. Equivalently, the logarithmic derivative is $\frac{d}{d\alpha} \log R_{\mathcal{P}} = -\sum_{p \in \mathcal{P}} 2p^\sigma/(p^{2\sigma} - \alpha^2) < 0$. For $\sigma > 1$, $\sum_p 2p^\sigma/(p^{2\sigma} - \alpha^2)$ converges uniformly on compact subsets of $[0, 1)$, so the infinite product inherits strict decrease. For $\sigma \le 1$, $\sum_p 1/p^\sigma = \infty$ implies $\log R_\infty = -\infty$, giving $R_\infty = 0$. □

Theorem 11.12 (Derivative of finite cancellation ratio)**.** *Write $T(N, \alpha) = \sum_{r \ge 0}(-\alpha)^r T_r$ and $D(N, \alpha) = \sum_{r \ge 0} \alpha^r T_r$ with $T_r = \sum_{\substack{k \le N,\, \mu(k) \ne 0 \\ \omega(k) = r}} k^{3\sigma} \lfloor N/k \rfloor > 0$. Define the probability weights $p_r = \alpha^r T_r / D$ and partition $\{r\}$ into even (E) and odd (O) subsets:*

$P_E = \sum_{r\,\text{even}} p_r$, $\quad P_O = \sum_{r\,\text{odd}} p_r$, $\quad \mu_E = \sum_{r\,\text{even}} r\, p_r/P_E$, $\quad \mu_O = \sum_{r\,\text{odd}} r\, p_r/P_O$. *Then the ratio* $\rho(\alpha) = -T(\alpha)/D(\alpha)$ *satisfies*

$$\frac{d\rho}{d\alpha} = -\frac{2}{\alpha}\, P_E\, P_O\, (\mu_E - \mu_O). \tag{12}$$

In particular, ρ *is strictly decreasing (cancellation* $\mathcal{C} = 1-\rho$ *is increasing) on any interval where* $\mu_E > \mu_O$.

Proof. Write $\rho = -T/D$ with $T < 0$ (verified numerically for $\alpha \geq 1/2$, proved via alternating series for large enough N). Using $T = \sum(-1)^r \alpha^r T_r$ and $D = \sum \alpha^r T_r$:

$$\begin{aligned} T'D - TD' &= \sum_{r,s} \big[(-1)^r r - (-1)^r s\big]\, \alpha^{r+s-1}\, T_r T_s \\ &= \sum_{r<s} \big[(-1)^r - (-1)^s\big](r-s)\, \alpha^{r+s-1}\, T_r T_s. \end{aligned}$$

Grouping by parity and simplifying using $P_E + P_O = 1$ gives $T'D - TD' = 2\,\alpha\, D^2\, P_E\, P_O\, (\mu_E - \mu_O)$, from which (12) follows. □

Proposition 11.13 (Log-concavity of T_r)**.** *For* $\sigma = 1/4$ *and* $N \leq 100{,}000$*, the sequence* T_r *is log-concave for* $r \geq 1$*:* $T_r^2 \geq T_{r-1}\, T_{r+1}$*. Moreover* $\mu_E > \mu_O$ *for all* $\alpha \in [1/2, 1]$*, so* $\mathcal{C}(N, \alpha)$ *is strictly increasing in* α *on* $[1/2, 1]$.

Verification. Computed for all $N \in \{1000, 5000, 10000, 20000, 50000, 100000\}$. The gap $\mu_E - \mu_O > 0$ decreases as $O(1/\log\log N)$, consistent with the Erdős–Kac theorem: both μ_E and μ_O converge to $\log\log N$, but μ_E retains a positive lead because T_2 (the dominant even block) exceeds T_1 (the dominant odd block). The log-concavity of T_r is expected from the Sathe–Selberg asymptotics for the distribution of $\omega(k)$ among squarefree integers. □

Remark 11.14 (Interpretation). Equation (12) reveals that $\omega = 2$ integers (products of two distinct primes) drive the cancellation most strongly: at $\alpha = 1$, the $\omega = 2$ block contributes over 85% of $\partial T/\partial\alpha$. Full multiplicativity ($\alpha = 1$) is the unique point where the eigenvalue cancellation is maximal and the zeros of ζ simultaneously emerge. Experiments up to $N = 100{,}000$ confirm $|T(N, 1)| < |T(N, \alpha)|$ for all $\alpha \in (0, 1)$ with zero violations.

Remark 11.15 (Arithmetic origin of the positivity). The α-deformation does not eliminate the main arithmetic obstacle. The generating function $F(z; N) = \prod_{p \leq N}(1 + z/p^{1/4})$, whose z^r-coefficient gives $A_r = \sum_{\omega(k)=r,\, \mu^2(k)=1} k^{-1/4}$, satisfies $F(-1)/F(1) \to 0$ exponentially, so the principal weighted Sathe–Selberg term is too balanced to force $\mu_E - \mu_O > 0$. The observed positivity is instead generated by the fractional-part error $E_r = -\sum k^{3\sigma}\{N/k\}$ in the decomposition $T_r = N\, A_r + E_r$. Thus the positivity lemma is governed by the same arithmetic oscillation that underlies Möbius sums and related RH criteria. This confirms that the α-monotonicity, while a genuine structural property, is a *reformulation* of the cancellation problem rather than a simplification.

11.6 Multiplicative variance reduction

The function μ is a group morphism from $(\mathbb{N}^*, \times)$ to $(\{+1, -1\}, \times)$, equivalently a morphism $(\mathbb{N}^*, \times) \to (\mathrm{GF}(2), +)$ via $k \mapsto (\omega(k) \bmod 2)$. This algebraic structure forces μ to cancel *better* than random signs.

Theorem 11.16 (Variance decomposition)**.** *For any $f : \{k \le N : \mu^2(k) = 1\} \to \{+1, -1\}$ and weights $w_k \in \mathbb{R}$:*

$$\Big(\sum_k f(k)\, w_k\Big)^2 = \sum_k w_k^2 \;+\; 2 \sum_{\substack{k<l \\ \mu^2(k)=\mu^2(l)=1}} f(k)\, f(l)\, w_k\, w_l. \tag{13}$$

If f is random ± 1 i.i.d., the cross term has expectation zero, giving $\mathbf{E}[S(f)^2] = \|w\|^2$. For $f = \mu$ (multiplicative), the cross term is systematically negative: numerical computation gives

$$\frac{\sum_{k<l} \mu(k)\mu(l)\, w_k w_l}{\|w\|^2} \in [-0.49,\ -0.35]$$

for $w_k = k^{3/4}\{N/k\}$ and $N \le 20{,}000$, corresponding to a variance reduction of 70%–97% compared to random signs.

Proof. Equation (13) is immediate from expanding the square. For random f: $\mathbf{E}[f(k)f(l)] = 0$ for $k \ne l$. The negativity of the cross term for μ is verified computationally; each prime p contributes negatively (the pairs k, l with $p \mid \gcd(k, l)$ have correlated signs that reduce the sum). □

Remark 11.17 (Connection to GF(2) and Sikidy)*.* The sign of $\mu(k) = (-1)^{\omega(k)}$ extracts the *parity bit* of the prime factorization. This is the same algebraic structure as the XOR operation in the Malagasy Sikidy divination system, where each figure is a vector in $\mathrm{GF}(2)^4$ and derivation rules correspond to addition in GF(2). The "doubly magic" squares of Sikidy — satisfying both classical sum and XOR conditions simultaneously — are an analogue of the dual additive/multiplicative structure that underpins the Riemann zeta function.

11.7 Dimensional invariance of rank

For $d \ge 2$, define the d-contracted GCD matrix by iterated summation: $S_0(h) = f_\sigma(h)$, $S_r(h) = \sum_{k=1}^{N} S_{r-1}(\gcd(h, k))$ for $r \ge 1$, and $T_N^{(d)}[m, n] = S_{d-2}(\gcd(m, n))$. For $d = 2$, this recovers T_N.

Theorem 11.18 (Dimensional invariance)**.** *For all $d \ge 2$: $\mathrm{rank}(T_N^{(d)}) = Q(N)$.*

Proof. We show by induction on r that $S_r(h)$ depends only on $\mathrm{rad}(h) := \prod_{p|h} p$ (the squarefree kernel of h).

Base case: $S_0(h) = f_\sigma(h) = \prod_{p|h}(1 - p^{3\sigma})$ depends only on the prime set of h, hence only on $\mathrm{rad}(h)$.

Inductive step: Assume $S_r(h)$ depends only on $\mathrm{rad}(h)$. For h and h' with $\mathrm{rad}(h) = \mathrm{rad}(h')$, the multisets $\{\gcd(h, k) : 1 \le k \le N\}$ and $\{\gcd(h', k) : 1 \le k \le N\}$ have the same image under rad. Since S_r depends only on the radical, $S_{r+1}(h) = \sum_k S_r(\gcd(h, k)) = \sum_k S_r(\gcd(h', k)) = S_{r+1}(h')$.

By Smith's theorem, the eigenvalues of $T_N^{(d)}$ are $(S_{d-2} * \mu)(k)$. For non-squarefree k, write $k = p^a \cdot m$ with $a \ge 2$ and $p \nmid m$. Every divisor e of k satisfies $\mathrm{rad}(k/e) = \mathrm{rad}(k/e)$, and since S_{d-2} depends only on the radical:

$$\begin{aligned}(S_{d-2} * \mu)(k) &= \sum_{e|k} S_{d-2}(k/e)\, \mu(e) \\ &= \sum_{e|k} S_{d-2}(\mathrm{rad}(k/e))\, \mu(e).\end{aligned}$$

Group the divisors by $\mathrm{rad}(k/e)$: for each squarefree r dividing $\mathrm{rad}(k)$, the contribution is $S_{d-2}(r)\sum_{e:\,\mathrm{rad}(k/e)=r}\mu(e)$. Since k has p^a with $a \geq 2$, the inner sum over e vanishes by the identity $\sum_{e|n}\mu(e) = [n = 1]$ applied to the p-part. Therefore $(S_{d-2} * \mu)(k) = 0$ for non-squarefree k, giving $\mathrm{rank}(T_N^{(d)}) = Q(N)$. □

Remark 11.19 (Structural universality). The squarefree integers are the *irreducible atoms* of the multiplicative structure at every level of interaction. The constant $Q(N)/N \to 6/\pi^2$ is a topological invariant of the divisor lattice: no increase in interaction dimension creates new atoms or destroys existing ones. This confirms that the $\sqrt{N}$ gap is intrinsic to the Möbius cancellation problem itself, not an artifact of the two-dimensional matrix formulation.

12 Connections to other approaches

12.1 Hilbert–Pólya

The Hilbert–Pólya conjecture seeks a self-adjoint operator with spectrum $\{\gamma_k\}$. We tested several candidates: the Möbius convolution operator T_μ (spectrum in $\{0, 1\}$, too simple), the Berry–Keating discretization (spectrum uniform, misses zeros), and our matrix $S_N(s)$ (symmetric for real s, detects zeros via $\sigma_{\min}$ but not through eigenvalue matching). Yakaboylu [13] proposes a Hamiltonian candidate whose self-adjointness remains unproven.

12.2 Connes' approach

Connes [14] uses noncommutative geometry on the adèle space. Our valuation space $V = \bigoplus_p \mathbb{N}$ is an "elementary" version of the adèlic decomposition $\mathbb{Q}_\mathbb{A} = \mathbb{R} \times \prod_p \mathbb{Q}_p$. The trace formula in our framework (trace $= \zeta(2s)/\zeta(-s)$) is a GCD-matrix avatar of Connes' trace formula.

13 Conclusion and open problems

Starting from the principle "the set of sets is not the same," we constructed a symmetry defect matrix $S(s)$ from divisor lattices and established:

- $\mathrm{trace}(S_N(s)) \to \zeta(2s)/\zeta(-s)$ for $\mathrm{Re}(s) > 1/2$ (Theorem 3.2);
- $\mathrm{rank}(S_N) = Q(N)$ independently of s (Theorem 3.4);
- $\|S_N(\sigma)\|_F^2 = \Theta(N^{2-4\sigma})$ unconditionally (Theorem 4.2);
- the trace decomposes into an unconditional pole term $N^{1-2\sigma}$ and oscillatory terms controlled by the zeros (Eq. 4);
- at $s = \rho/2$, the trace grows as $N^{1-\mathrm{Re}(\rho)}$: a zero off the critical line produces anomalously large growth (Theorem 4.1);
- $\sigma_{\min}(S_N(1/4 + it))$ collapses at $t \approx \gamma_k/2$, numerically detecting the first zeros of ζ (Section 8);

- the trace admits a Ramanujan decomposition with $(f_s * \mu)(d) = \mu(d)\, d^{3s}$ (Theorem 11.1);
- an exact Hadamard identity links $S_N(\sigma)$ to $S_N(1-\sigma)$ (Theorem 11.3), with R_N positive semi-definite (Proposition 11.4);
- numerical evidence suggests operator-norm transfer across the critical line (Remark 11.5);
- numerical cancellation of eigenvalue sums exceeds 99.7% at $N = 5000$ (Section 11.3).

The matrix S_N thus provides a *set-theoretic detector* for the nontrivial zeros: their real parts are encoded in the growth rate of the trace at the image point $s = \rho/2$. The Riemann Hypothesis is the assertion that this growth rate equals 1/2 for *every* nontrivial zero — equivalently, that no anomalous growth is observed.

The Hadamard identity (9) reveals that the two sides of the critical strip are linked by an exact entrywise relation. The positive semi-definiteness of the ratio matrix R_N enables operator-norm transfer, but a gap of $\sqrt{N}$ remains between operator-norm bounds and trace bounds. This gap is the same obstacle encountered in all known approaches to RH: it amounts to proving that the multiplicative structure of μ forces sufficient sign cancellation among the $Q(N)$ eigenvalues.

The numerical prediction $\gamma(\sigma) = 2 - 4\sigma$ (derived from the unconditional Frobenius norm and assuming bounded trace) matches the measured exponent with correlation 0.95 for $\sigma > 0.30$ (Section 8), providing quantitative evidence that the trace does not exhibit anomalous growth in this range.

Four directions remain open:

1. **Closing the $\sqrt{N}$ gap:** the Hadamard identity transfers operator norms across the critical line, but the trace involves all $Q(N)$ eigenvalues, not just the extremal one. Closing this gap requires proving that the 50/50 sign balance of $\mu(k)$ forces cancellation at the $O(\sqrt{N})$ level. This is equivalent to RH via classical results (Littlewood), but may admit a proof through the matrix structure.

2. **Exploiting the Hadamard identity:** the exact relation $S_N(1-\sigma) = D^{-1}(S_N(\sigma) \circ R_N)D^{-1}$ has not been fully exploited. In particular, inequalities involving the *off-diagonal* entries (not just the trace) may yield non-trivial constraints. The positive semi-definiteness of R_N enables Schur-type bounds that could transfer cancellation information across the critical line.

3. **Spectral cancellation via multiplicativity:** the eigenvalue signs $\mu(k) = (-1)^{\omega(k)}$ are not independent; they satisfy $\mu(mn) = \mu(m)\mu(n)$ for coprime m, n. This multiplicative constraint reduces the variance of the eigenvalue sum by a factor exceeding 12 at $N = 2000$ compared to independent random signs. Quantifying this reduction rigorously would yield RH.

4. **Matrix functional equation:** construct a "completed" matrix $\Xi_N(s)$ (analogous to $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$) satisfying $\Xi_N(s) = \Xi_N(1-s)$ spectrally. The GCD matrix $\Xi[m,n] = \pi^{-s/2}\Gamma(s/2)\,\gcd(m,n)^{-s}$ has this symmetry at the trace level but does not encode the zeros of ζ. Finding a matrix that has both the symmetry and the zero information remains open.

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